

Krish Patel

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EDUCATION

University of California, Irvine | Irvine, CA

Sep 2025 – Present

Bachelor of Science in Electrical Engineering, Henry Samueli School of Engineering | GPA: 3.33/4.0

Spring Valley High School | Columbia, SC

Aug 2021 – Jun 2025

High School Diploma, Discovery Research Magnet Program | GPA: 3.91/4.0

TECHNICAL SKILLS

- **Programming:** C/C++, Python, Java, Verilog, MATLAB
- **Machine Learning & Data Analysis:** Reinforcement Learning, Q-Learning, Matplotlib, ImageJ
- **Electronics & Embedded Systems:** Circuit Design, PCB Design, KiCad, Altium, Arduino, Microcontrollers, Soldering
- **Mechanical Design:** CAD, Autodesk Fusion
- **Systems & Lab:** System Integration, Systems Tool Kit (STK), Microbiology

RESEARCH

Integrated Quantum Photonics | [Shcherbakov Nanophotonics Lab](#), UC Irvine

Sep 2026 – Present

- Explored the integration of photonic hardware into quantum computing architectures, focusing on the use of on-chip optical components as a scalable alternative to traditional free-space laser setups in trapped-ion systems.
- Studied the replacement of bulky free-space laser optics with chip-scale photonic platforms, leveraging microfabricated ion traps paired with integrated photonic circuits for more compact and stable qubit control.
- Investigated integrated photonic components — including on-chip waveguides, grating couplers, and electro-optic modulators — designed to route and deliver laser light directly to trapped ions with high precision.
- Analyzed performance parameters of grating couplers and waveguide structures for efficient photon delivery to trapped ion qubits, with emphasis on coupling efficiency, mode overlap, and fabrication compatibility with ion trap processes.

PROJECTS

Systems Engineering Subteam | UC Irvine CubeSat

Aug 2026 – Present

- Supported systems engineering across a full CubeSat development cycle, ensuring all subsystems integrated successfully to meet overall mission requirements.
- Managed integration across avionics, communications, and power teams while strictly tracking mass, power, and link budgets.
- Developed core technical documentation, including Concept of Operations (CONOPS) diagrams and Interface Control Documents (ICDs).
- Validated orbital paths via Systems Tool Kit (STK) and executed hardware-software verification on a FlatSat test bench.
- Collaborated with the Power sub-team to verify power distribution and manage current draw allocation.

Microcontroller-Based Oscilloscope | github.com/krishpatelkp2423-dot/Oscilloscope

Aug 2026 – Sep 2026

- Designed a general-purpose oscilloscope from scratch to explore how analog front-end design affects a microcontroller's usable ADC resolution.
- Compared three architectures on one board: direct ADC sampling, hardware auto-ranging, and an external 16-bit ADC.
- Engineered an auto-ranging stage that level-shifts the input to 0–3.3 V, then uses window comparators to detect voltage bands and gain stages to fill the ADC's full range.
- Laid out a custom KiCad PCB, hand-soldered the board, and wrote a Python app for real-time waveform visualization.

Machine Learning Chess Model | <https://github.com/krishpatelkp2423-dot/Chess>

Mar 2026 – Apr 2026

- Built a playable chess variant in Python (Pygame) that replaces checkmate with a point-based win condition (first to 15).
- Reworked core mechanics so each captured piece carries a point value, with a real-time side-panel scoreboard that shifts strategy toward risk-reward tradeoffs.
- Trained a reinforcement learning agent (Q-learning/policy-based) to learn the modified ruleset from scratch.
- Compared the agent's adaptation with human strategies to study how each handled unfamiliar incentive systems.

Autonomous Mobile Rover | Students for the Exploration and Development of Space, UC Irvine

Mar 2026 – Apr 2026

- Built a rover that autonomously navigates simulated extraterrestrial terrain, locates a target rock, and collects and stores it.
- Combined CAD-designed mechanics, electronics, and embedded software into one integrated platform under tight power and computing limits.
- Integrated mechanical, electrical, and software subsystems, troubleshooting interfaces to improve reliability.
- Designed circuit schematics for an Arduino (C/C++) control system with DC motors, motor drivers, sensors, and power management.